

Dinh Viet Luong

UWaterloo Electrical Engineering Student

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in Dinh Viet Luong 📌 DINDIN2007

Skills

- **Programming Languages:** C++, Java, Kotlin, Swift, Python, Bash, HTML, CSS, JS, XML, Verilog, QML
- **Developer Tools:** Altium Designer, KiCad, Arduino CLI, SolidWorks, SketchUp, VS Code, Eclipse, Android Studio, XCode, Vim, LaTeX, Microsoft Office 365, Excel, Git, Notion, OS (Windows, macOS, Linux, Virtual Machines)
- **Networking Protocols:** Bluetooth Low Energy, Generic Attribute Profile, UART, SPI, I2C, SSH
- **Hardware:** Soldering Circuitry, Multimeter, Oscilloscope, Arm® Cortex™-M4, nRF52832, nRF52840, nRF54L15, ICM-20948, LSM6DSV80X, nPM1304 PMIC, TMP117, MAX30102, SSD1306, Arm® Cortex Debugger, Arduino UNO, RPi4
- **Languages:** English, French, Vietnamese, Spanish, Japanese

Work Experience

IoT Engineering Intern

I-Syst.inc

Boucherville, QC

July 2024 - Aug 2024, Jan 2026 – May 2026

App Demo for ICM-20948 Sensor on BLE Board

github.com/I-SYST/AppDemo 🔗

- Developed an **Android Application** (primarily in **Java**) to connect and acquire real-time data from a custom **BLE** device, which was used to track and record the force and acceleration motion of a boxing athlete during their workout.
- Implemented a hardware-to-software interface using BLE **Generic Attribute Profile (GATT)** Connection Protocol to perform service discovery, configure sensor characteristics, and subscribe to notifications for real-time data streaming.
- Processed raw motion data into dynamic charts and saved it into a csv file for analysis in external software (ex. Excel).

Smart Menstruation Tracker and Heatpad Controller

- **Tools Used:** Android Studio (Kotlin, XML), XCode (Swift), Bluetooth Low Energy, Machine Learning
- Engineered a cross-platform **BLE** mobile application that connects and provides temperature control for smart heatpad.
- Developed a lightweight predictive tracking engine that leverages a **Recurrent Neural Network** and implements an **Exponential Moving Average** fallback mechanism which adjusts based on past user data deviations.
- Designed an intuitive user interface featuring an unobtrusive, predictive notification system tailored to the user's cycles.

Cross Platforms Installation Scripts

github.com/IOsonata/IOsonata 🔗

- **Tools Used:** Bash Scripting, Powershell Scripting
- Authored automated installation scripts for proprietary firmware dev tools, which accelerates the onboarding process.
- Consolidated a deployment of Eclipse IDE, custom AI plugins, and toolchain dependencies into a single terminal command for all three main OS, as well as automating preference seeding to deliver a zero-configuration, ready-to-code environment.
- Included an automated development environment removal and streamlined project/library building for Nordic BLE SoCs.

Modernizing Bluetooth Low Energy Mobile Applications

bit.ly/I-SYST-Apps 🔗

- Modernized five legacy apps, upgrading core dependencies and ensuring compliance with the latest BLE requirements.
- Standardized the UI/UX across the app ecosystem to enhance cross-app navigation and establish a cohesive brand identity.
- Refactored BLE communication protocols, which reduced application crash rates and establishing stable device connections.

Projects

Find out more on my website: dindin2007.github.io/Portfolio/ 🔗

Custom Smartwatch PCB Design

github.com/DINDIN2007/Smart-Watch-L15 🔗

- Architected and routed a custom smartwatch and docking station, integrating the ultra-low-power **nRF54L15 SoC** with an NFC tag and modern sensors (including the LSM6DSV80X) for a highly resource-constrained embedded environment.
- Established **SPI** and **I2C** sensor communication, and presently developing the C/C++ firmware for data acquisition.
- Designed the **PCB** to meet the space constraints of a wearable, such as thermal isolation and antenna clearance zones.
- Currently modeling the smartwatch case and docking station utilizing **SolidWorks** for precision CAD integration.

Integrated Circuit Flash Breakout PCB for FPGA Module

github.com/uwrealitylabs/HD3SS3220-breakout 

- Architected full-system schematics and component selection for a 4-layer PCB bridging **USB 3.0 Type-A** and **Type-C** (with the HD3SS3220 port controller in between), enabling practical physical-layer validation of an FPGA implementation.
- Created the **PCB** layouts in **Altium** and **KiCad** and grouped high-speed data wires into pairs and matched their impedance to exactly 90 ohms for the 5Gbps SuperSpeed signals to travel across the board without losing any information.
- Designed a clean layout for the Electrostatic Discharge (ESD) protection chips, ensuring that sudden spikes of static electricity are safely directed away from the main circuitry to prevent permanent hardware damage.

Active Noise Cancellation System

github.com/Ron12777/198 

- Designed and implemented a real-time Active Noise Cancellation system on a **Raspberry Pi** for a university project.
- Built an **adaptive cancellation loop** that utilized reference + error mics to continuously cancel out noise at a distance.
- Reduced audio latency by perfectly synchronizing the physical mic inputs and speakers with the **MATLAB** software.

Internationally Focused C++ Axmol Engine Game

github.com/Karlosian/ContinentalBuffet 

- Created a video game using a fork of Cocos2d-x with primarily **C++** that teaches famous dishes from all over the world.
- Designed an object-oriented system to parse and load recipe data from custom **json** files, allowing codeless content updates.
- Utilized **Git** for version control in a collaborative team setting, enabling parallel workflows and seamless code integration.

Productivity Desktop User Interface

github.com/DINDIN2007/Thinkers-Desktop-Bar 

- Engineered a custom, interactive desktop environment using **Quickshell**, replacing static status bars with modular widgets.
- Integrated **Bash scripts** with the Hyprland to automate system updates, monitor hardware stats, and manage workspaces.
- Designed a scalable and minimal UI architecture to organize app launchers, quick settings, and clipboard history.

Education

University of Waterloo

Sept 2025 – Present

Bachelor's in Engineering

- Obtained the President's Scholarship of Distinction and is expected to graduate in April 2030.
- **Courses:** Digital Circuits and Systems, Linear Circuits, Electricity and Magnetism, Fundamentals of Programming, ...

Certificate

Communication Series P1: UART, SPI and I2C in Verilog

Apr 2026

Udemy Certificate of Completion

- Finished introductory course on principles of digital communication interfaces while employing Verilog as the primary tool.
- Certificate no: UC-1a7a7f2c-f2e6-4096-a2e0-96eee3290c56

Learn Linux Kernel Programming

Mar 2026

Udemy Certificate of Completion

- Finished introductory course on how to write Linux Kernel Modules and Device Drivers.
- Certificate no: UC-c64b6dde-3098-4a6a-857d-0783eba3ccdd

KAKEHASHI Project Alumni

Feb 2026

Asia Pacific Foundation of Canada

- Government-sponsored exchange program that promotes understanding of Japan and strengthens international relations.

French Education International

May 2025

Delf B2 Certificate of Achievement

- Passed the DELF B2 exam with an overall score of 86% in Ontario high school's French immersion stream.

International Certificate Program

Sept 2021 – June 2025

Honours Diploma

- Participated in international-focused events, welcomed international students, and organized cultural exchange activities.